NexusOS
AI Business Operating System

Part 03.0
Linear Issue WAL-12

Migration Development

Project	NexusOS — AI Business Operating System
Part	Part 03.0
Linear issue	WAL-12
Report title	Migration Development
Generated	2026-07-19
Purpose	Standalone Linear attachment (manual upload)
Source report(s)	docs/reports/Part-03.0/Part-03.0-Migration-Development.md

This PDF is a presentation export of repository engineering report content. Source report files in the repository were not modified.

Evidence Statement

This report is a retrospective engineering reconstruction produced after implementation. Its contents are derived exclusively from repository evidence. It is not a contemporaneous engineering document. No engineering events are fabricated. Where repository evidence is unavailable, that limitation is explicitly documented.

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Report Content

Content below is copied from the source repository report(s) without alteration.

Part 03.0 — Migration Development

Linear: WAL-12 **Reconstruction date:** 2026-07-19

Evidence Statement

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Objective

Establish an Alembic migration baseline so the enterprise schema is versioned and upgradeable from an empty database.

What was implemented

Initial revision

- **Revision ID:** `0a961c2ea2cf`
- **File:** `backend/migrations/versions/20260709_0a961c2ea2cf_initial_schema.py`
- **Create Date (in file):** 2026-07-09 11:31:26
- **down_revision:** `None` (root)
- **Tables created:** 37
- **Indexes created:** 150 (`op.create_index` count)

Tooling

- `backend/alembic.ini`, `backend/migrations/env.py`
- Makefile targets documented in `database.md`: `make revision`, `make migrate`

Later chain (not Part 03.0 authorship, but dependency context)

Revision	Part	Purpose
`382d8497298e`	02	Sessions and auth tokens
`3a2aec52f6d2`	03.1	Fix `created_by/updated_by` FK drift
`217e90d00cc0`	03.1	Composite tenant FKs + soft-delete-safe uniques

Repository evidence

- Initial migration file above
- `backend/migrations/env.py`, `alembic.ini`
- Commit `838db64`
- `docs/product/database.md` working-with-schema section

Validation

Part 03.1 reports later document from-empty upgrade and round-trip validation of the full chain including this root revision. No separate contemporaneous 03.0 migration validation transcript is stored under `docs/reports/Part-03.0/` prior to this reconstruction.

Outcome

Reliable Alembic root for the enterprise schema.

Evidence limitations (shared)

- Part 03.0 code primarily landed in git commit `838db64` (2026-07-09) under the commit message “chore: validate and stabilize backend foundation”, bundled with broader foundation scaffolding. There is no separate contemporaneous commit uniquely titled “Part 03.0”.
- Linear issues WAL-7…WAL-17 were created on 2026-07-18 as a historical execution hierarchy after the schema already existed in the repository.
- No contemporaneous Part-03.0 independent-review finding IDs, meeting notes, or signed review PDF exist in the repository (unlike Part 03.1’s report pack).
- Structural integrity remediations (composite cross-tenant FKs, soft-delete-safe unique indexes, missing `created_by`/`updated_by` FKs) are **Part 03.1** evidence (`f521d38`, revisions `3a2aec52f6d2` / `217e90d00cc0`) and are not attributed as silent Part 03.0 remediations.